

Anjali Tiwari

Safe Robot Planning and Control under Uncertainty · Optimization · Learning-Augmented Control

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RESEARCH PROFILE

I am a researcher working in the domain of optimization-based controllers that give robots formal safety and stability guarantees under model uncertainty, disturbances and limited sensing. Methods: CBF/CLF quadratic programs, HJB-based optimal control, discrete-time barrier functions, static output feedback and RL-augmented control. Applications: single- and multi-robot navigation among static and dynamic obstacles under bounded disturbances and unknown flow fields, and manipulator safety under uncertainty. Next goal: validate these guarantees on physical multi-robot teams.

EDUCATION

PhD, Control Systems and Automation — Indian Institute of Technology Jodhpur, India 2020—2026

Thesis Title: *Unifying Safety and Robustness for Control of Dynamical Systems under Uncertainty*. (Submitted 2026)

Supervisor: Dr. Niladri Sekhar Tripathy.

8.25/10.

M.Tech, Electrical Engineering — National Institute of Technology Hamirpur, India 2018—2020

Thesis Title: *Synchronization of Chaotic and Hyperchaotic Systems using Sliding Mode Control*.

Supervisor: Dr. Himesh Handa.

8.26/10.

B.E., Electrical Engineering — Ujjain Engineering College, India 2013—2017

8.11/10.

CURRENT POSITION

Research Associate — Department of Electrical Engineering, IIT Jodhpur 2026—present

RESEARCH EXPERIENCE

Safe robot navigation under uncertainty via optimization and learning 2026—present

- Designed a unified RL + optimization controller for navigation in uncertain, obstacle-cluttered environments, with CBF/CLF constraints enforced in a real-time QP safety layer.
- Developed safe robust obstacle avoidance and multi-obstacle navigation for a unicycle mobile robot under bounded disturbances, including a learning-augmented CLF—CBF controller (accepted, ICC).
- Extended safe navigation to multi-agent teams: learning-based go-and-return and robust circumnavigation among dynamic obstacles under unknown, nonuniform flow fields (two papers submitted to ACC).
- Applied barrier Lyapunov control beyond robotics, to enforce a strict safety margin in LLM text generation (accepted, ICC).

Safe robust static output-feedback control 2025—2026

- Guaranteed safety and stability using measured outputs only, by embedding safety as discrete barrier states in the system dynamics; suited to robots that act on partial or estimated state.

Safety-embedded discrete-time optimal control 2023—2025

- Embedded discrete-time CBF constraints in the optimal-control cost, unifying safety, stability and optimality without constraint tightening; validated on a single-link manipulator under bounded uncertainty.

Safe robust optimal control of uncertain nonlinear systems (PhD core) 2021—2023

- Recast robust safe control as constrained optimal control solved via the HJB equation and KKT conditions, with a switching strategy that preserves safety; covers matched and mismatched uncertainty.

PUBLICATIONS

Journal articles

- [1] **A. Tiwari**, N. S. Tripathy, M. Chamanbaz, R. Bouffanais, “Ensuring safe robust control design through optimal control with switching strategy,” *Asian Journal of Control*, 2026.

- [2] **A. Tiwari**, N. S. Tripathy, "A unified framework for robust stabilization and safety of uncertain nonlinear systems," *ASME Journal of Dynamic Systems, Measurement, and Control*, 148(5), 2026.

Conference papers (accepted)

- [3] **A. Tiwari***, S. Singh*, A. Singh, "Safe unicycle navigation among multiple obstacles under bounded disturbances: a learning-augmented CLF–CBF approach," *Indian Control Conference (ICC)*, accepted.
- [4] A. Singh, S. Singh, **A. Tiwari**, "Enforcing a strict safety margin in LLM text generation via barrier Lyapunov control," *Indian Control Conference (ICC)*, accepted.

Submitted

- [5] **A. Tiwari**, S. Singh, A. Singh, Y. Wang, K. Fujisawa, "Safe learning-based multi-agent go-and-return among dynamic obstacles under unknown flow," submitted to *American Control Conference (ACC)*.
- [6] S. Singh, **A. Tiwari**, A. Singh, Y. Wang, K. Fujisawa, "Robust safe circumnavigation among dynamic obstacles under unknown nonuniform flow fields," submitted to *American Control Conference (ACC)*.
- [7] **A. Tiwari**, N. S. Tripathy, "Safety-embedded robust static output feedback control for discrete-time systems under unmatched uncertainty," under preparation.

* Equal contribution.

TECHNICAL SKILLS

- **Planning and control:** CBF/CLF-QP safety filters, discrete-time barrier functions and barrier states, obstacle avoidance and multi-agent navigation among dynamic obstacles, HJB-based optimal control, robust and output-feedback control, RL-augmented control.
- **Optimization:** quadratic programming, KKT-based constrained optimal control, real-time QP formulation.
- **Programming:** Python, MATLAB/Simulink, C++ (basic), Verilog HDL.
- **Tools:** LaTeX, Ipe, FPGA.

COLLABORATIONS

IIT Jodhpur (India) · Rio Tinto Centre for Mine Automation, University of Sydney (Australia) · University of Geneva (Switzerland)

TEACHING AND SERVICE

- Teaching Assistant, IIT Jodhpur, 2020–2025: Control Systems, Cyber-Physical Systems, Advanced Control.
- Peer reviewer: *ASME Journal of Dynamic Systems, Measurement, and Control*; Indian Control Conference; Advances in Robotics.

AWARDS AND FELLOWSHIPS

- MHRD Junior/Senior Research Fellowship, Government of India, 2020–2025.
- MHRD Postgraduate Scholarship, Government of India, 2018–2020.
- GATE qualified (Electrical Engineering), 2018.

ACADEMIC AND PROFESSIONAL ACTIVITIES

Master of Ceremonies, Advances in Robotics (AIR 2025)	2025
7th International Conference of the Robotics Society; hosted technical sessions and ceremonial events.	
Workshop on Observer and Control Design for Autonomous Vehicles, IIT Bombay	2024
Workshop on Field Robotics, Autonomous Systems and Robotic Applications, IIT Jodhpur	2023
Two-day workshop.	

REFEREES

Dr. Niladri Sekhar Tripathy — Assistant Professor, Department of Electrical Engineering, IIT Jodhpur, India, PhD supervisor, niladri@iitj.ac.in, +91 291 280 1375

Prof. Bijnan Bandyopadhyay — Distinguished Professor, School of Computing and Electrical Engineering, IIT Mandi, India; Life Fellow, IEEE, Teaching-assistantship supervisor at IIT Jodhpur, bijnan@iitmandi.ac.in